Aishwarya Sharma

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EDUCATION

International Institute of Information Technology, Bangalore **August 2023 - Present**
5 year Integrated Masters of Technology in Computer Science and Engineering *CGPA: 3.35/4.00*

EXPERIENCE

Research Intern – International Institute of Information Technology Bangalore **May 2026 – Present**
Semantic Firewall: AI-Driven Security Analysis Framework *Onsite*

- Proposed a semantic analysis framework for cybersecurity applications, utilizing machine learning and program analysis techniques to identify malicious patterns beyond traditional signature-based detection methods.
- Leveraged feature extraction and behavioral modeling to characterize software artifacts and evaluate semantic-aware threat detection approaches against obfuscation and code transformation techniques.

Teaching Assistant – Computer Architecture **January 2026 – May 2026**
Mentorship Role *IIIT Bangalore*

- Mentored 160+ students in Computer Architecture through lab sessions and tutorials, providing guidance on assembly programming, pipelining, cache memory, and performance analysis.

PROJECTS

Store-Delivery Management System | *Java, SQL, JDBC* ***May 2025***

- Integrated Java with SQL using JDBC to build a command-line Store–Delivery Management System, enabling efficient management of inventory, orders, and delivery workflows through transaction-safe CRUD operations and optimized database interactions.
- Structured a normalized relational database schema and applied the MVC architectural pattern to improve data consistency, scalability, and maintainability, with comprehensive exception handling to enhance system reliability.
- Developed and validated an interactive CLI interface for real-time data insertion, updates, and retrieval, verified through unit testing with representative sample datasets.

GramUdyogAI (Empower Up) | *Python, React, TypeScript, Tailwind CSS, FastAPI, NLP* ***April 2025 - July 2025***

- Engineered an AI-driven rural entrepreneurship platform delivering personalized business ideas, multilingual assistance, and access to government schemes, optimized for low-literacy and low-connectivity environments.
- Implemented AI-generated business recommendations, regional-language audio summaries of YouTube business content, and a localized job board to bridge the digital divide for underserved rural communities.
- Architected and deployed a scalable full-stack system using React (TypeScript, Tailwind CSS) and FastAPI (Python), supporting **10+ regional languages**, reducing content consumption time by **~40%**, and improving engagement for **1,000+ early users** via lightweight UI and cloud-hosted APIs on Railway and Vercel.

ETL Pipeline Analysis | *Python, MongoDB, ETL Pipeline, Data Analysis* ***May 2026 – July 2026***

- Designed and implemented an end-to-end ETL pipeline to extract, transform, and load large-scale structured and semi-structured datasets into a MongoDB-based NoSQL database.
- Developed automated data-cleaning and transformation workflows using Python, enabling efficient preprocessing, schema validation, and integration of heterogeneous data sources.
- Performed exploratory data analysis and generated interactive visualizations to uncover trends, optimize query performance, and support data-driven decision making.

ACHIEVEMENTS

Top 10 Finalist – Meta Hackathon | *National-Level Competition* *July 2025*

- Secured a position among the **Top 10 Finalists** in the **Pragti AI Meta Hackathon**, recognized for developing an innovative AI-powered solution addressing real-world social and technological challenges.

TECHNICAL SKILLS

Languages: Python, SQL, C++, Java, C, JavaScript
Libraries/Tools/Frameworks: Pandas, Numpy, Git, GitHub, React.js, Vercel, Railway, Figma

RELEVANT COURSEWORK

Coursework: Data Structures, Algorithms, Object-Oriented Programming, Database Management Systems, Computer Networks, Operating Systems, Machine Learning, Programming Languages, NoSQL Databases

EXTRACURRICULAR

Parvaaz - IIITB Theatre Club | *Core Member* *January 2024 - present*

- Collaborated with a 10+ member team to conceptualize, script, and perform stage productions, enhancing creative storytelling and coordination skills.